Project Design Phase
Solution Architecture

Date	14 October 2026
Team ID	
Project Name	Metro Ticket Generating System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram – Metro Ticket Generating System:

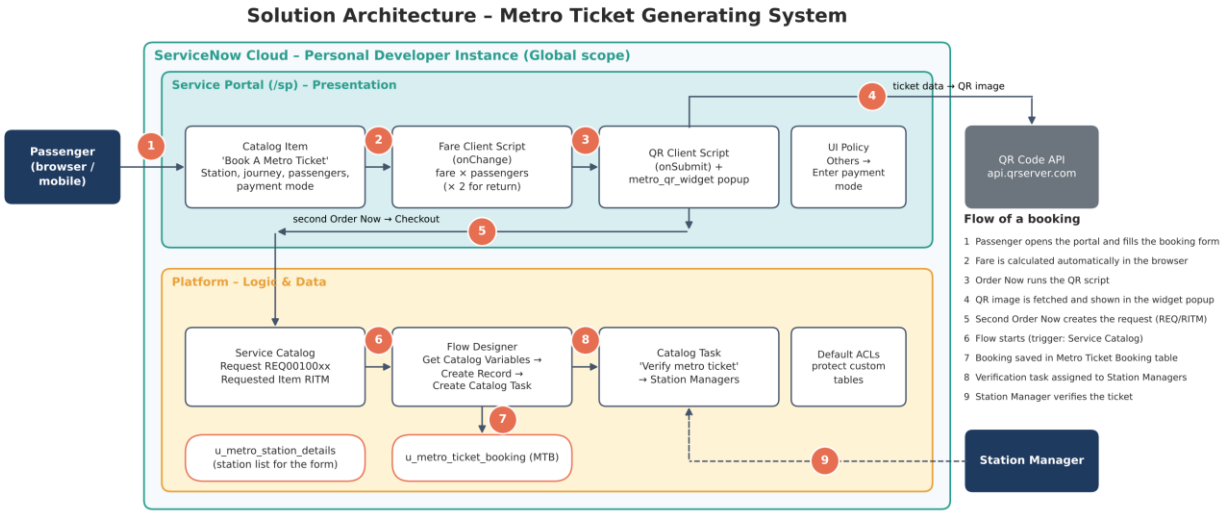


Figure 1: Architecture and data flow of the Metro Ticket Generating System on ServiceNow

Explanation of the architecture:

• Presentation (Service Portal): the passenger uses the catalog item 'Book A Metro Ticket' with 8 variables. Station lists come from the Metro Station's Details table.

• Fare logic: an onChange catalog client script on No of Passengers calculates the amount (route fare × passengers, doubled for a return journey) and fills a read-only amount field.

• Payment rule: a catalog UI policy shows the mandatory 'Enter payment mode' field only when Mode of Payment is Others.

• QR ticket: an onSubmit catalog client script sends the ticket data to the QR Code API and opens the metro_qr_widget popup. The flag g_form.metroQrDone lets the second Order Now place the order.

• Request: the Service Catalog creates a request (REQ) and requested item (RITM) that the passenger can track in My Requests.

• Automation: Metro Ticket Booking Flow (Flow Designer, run as System) gets the catalog variables, creates a record in the Metro Ticket Booking table and creates a catalog task for the Station Managers group.

• Security: default ACLs on both custom tables stop normal users from opening the tables directly.